A New Detection Method for Land Surface Anomalies From the Perspective of Thermal Infrared Remote Sensing

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***Abstract*—On-orbit rapid detection of land surface anomalies is important for ensuring ecological security and human safety. Land surface anomalies (e.g., fire, industrial heat source, deforestation etc.) are often accompanied by different degrees of thermal anomalies. Existing methods for detecting thermal anomalies have focused primarily on high-temperature anomalies, without an available approach for detecting widespread low-temperature anomalies. Here, a novel method based on constructed reference land surface temperatures (LSTs) for on-orbit remote sensing detection of various thermal anomalies (NMCRTA) is proposed and further evaluated using the Landsat 8 LST product. In this method, we first construct a fit reference temperature based on LST spatiotemporal trend surface modeling and a real reference temperature based on contextual averaging. Then, the difference (including the step of removing atmospheric effects) between on-orbit observed LST and fit reference LST, and the difference between on-orbit observed LST and real reference LST are calculated, respectively. Finally, these two temperature differences are utilized to detect thermal anomalies using corresponding thresholds. The results indicate that the NMCRTA can effectively detect deforestation, newly constructed buildings, and river drying, with an overall *F*1-Score of 0.838, in a 100 × 100 km region scale. Meanwhile, the NMCRTA exhibited excellent accuracy in detecting fires, deforestation, and landslides at the 15 × 15 km scene scale, achieving *F*1-scores of 0.943, 0.857, and 0.791, respectively. Furthermore, the NMCRTA can continuously capture different thermal anomaly events associated with a newly constructed industrial heat source and perform well at nighttime. The NMCRTA is promising for the future on-orbit remote sensing detection of various land surface anomalies, as a valuable supplement to optical on-orbit detection methods.**

***Index Terms*—Fit reference temperature, land surface temperature (LST), real reference temperature, thermal anomaly.**

Abbreviation and Symbol

LST	Land surface temperature.
NMCRTA	Novel method based on constructed reference LSTs for on-orbit remote sensing detection of various thermal anomalies.
STSLST	Spatiotemporal trend surface of the LST.
HR	High resolution.
ATC	Annual temperature cycle.
STD	Standard deviation.
DOY	Day of the year.
T_{sat}	Observed LST from satellites.
$T_{\text{solarangle}}$	Fit reference LST (i.e., STSLST).
T_{context}	Real reference LST.
ΔT_{atm}	Influence of atmospheric conditions.
$\Delta T^{\text{STSLST}}_{\text{remove_atm}}$	Atmosphere-removed temperature difference between the observed and fit reference LST.
$\Delta T^{\text{context}}_{\text{without_atm}}$	Without-atmosphere temperature difference between the observed and real reference LST.
MAST	Mean of the annual LST.
YAST	Amplitude of the annual LST.
dx	Phase shift of the annual LST.
T_α	Threshold of $\Delta T^{\text{STSLST}}_{\text{remove_atm}}$.
T_β	Threshold of $\Delta T^{\text{context}-}_{\text{without_atm}}$.

Received 10 February 2025; revised 30 June 2025; accepted 27 August 2025. Date of publication 2 September 2025; date of current version 16 September 2025. This work was supported by the National Natural Science Foundation of China under Grant 42192580. *(Corresponding author: Biao Cao.)*

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Digital Object Identifier 10.1109/TGRS.2025.3605164

I. Introduction

VARIOUS surface anomalies, including natural disasters, environmental pollution, and ecological destruction, have occurred frequently on Earth with the intensification of global climate change and human activity [1], [2]. These anomalous events have serious impacts on the natural environment, socioeconomic development, and public health [3], [4], [5]. Therefore, rapid detection of land surface anomaly events is of great importance to mitigate the impacts [6], [7]. Owing to the wide imaging coverage and low cost, remote sensing has become the primary method for detecting land surface anomalies [8]. The existing detection methods primarily rely on the changes in spectral reflectance, radar backscattering coefficients, or LST [9], [10], [11]. For example, deforestation is often detected using multitemporal optical images [12], landslides are typically detected using multitemporal radar images [13], [14], and volcanic activity is usually monitored with multitemporal thermal infrared images [15]. These methods can effectively identify these anomalies by comparing the before-anomaly and after-anomaly images. However, obtaining postevent images requires a lengthy process (including task planning, satellite operation control, ground data transmission, and data processing), which typically takes 1–4 days [16], [17].

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These methods cannot meet the real-time detection requirements for some sudden anomalous events, such as wildfire, heatwaves, factory explosions, landslides [6], [18], [19], [20], [21], [22].

In recent years, on-orbit detection utilizing a single-temporal remote sensing image has received increasing attention and is seen as a primary way to achieve real-time detection of land surface anomalies [23], [24], [25]. The detection process can be greatly simplified by directly deploying lightweight algorithms on satellites. Until now, a few on-orbit remote sensing detection methods for specific land surface anomalies have been proposed, such as fire detection and water quality anomaly detection [17], [26]. The on-orbit detection method for various land surface anomalies is urgently needed, which could be developed according to different radiative transfer principles: reflectance-based method, radar-based method, and LST-based method [27]. For that, Wei et al. [28] proposed a lightweight multidimensional feature space method based on reflectance data, which can identify floods, red tides, vegetation pests and diseases, and building collapses. Due to the complex processing requirements of radar-based methods, they are now difficult to adapt to the limitations of weak satellite computing capabilities [29]. Therefore, it is valuable to develop a lightweight detection method from the perspective of thermal infrared, as a supplementary approach of current optical on-orbit remote sensing anomaly detection.

In the field of thermal infrared remote sensing, various methods have been developed to detect LST anomalies using single-temporal remote sensing images (i.e., the land surface thermal anomalies), which can be divided into two categories: spectral-based and spatial-based. The spectral-based method detects thermal anomalies based on Wien's law of displacement that the peak of radiant emission shifts to shorter wavelengths as the temperature of an object increases [30], [31]. It is widely applied in fire and industrial heat source detection, which is effective for temperatures above 400 K [30], [32]; however, it cannot detect low-temperature thermal anomalies. The spatial-based method detects thermal anomaly by comparing the LST difference between the central and neighboring pixels with a determined threshold [33], [34]. It is unstable and unreliable for detecting low-temperature thermal anomalies (such as small fires and deforestation) due to spatial heterogeneity, geographical location, weather, and other factors [7]. Therefore, it is challenging to achieve on-orbit real-time detection of both high-temperature and low-temperature thermal anomalies.

In this study, an NMCRTA is proposed. This method comprehensively utilizes the difference between on-orbit observed LST and fit reference LST, and the difference between on-orbit observed LST and real reference LST. Section II introduces the related data, thermal anomaly detection method, and accuracy evaluation method. Sections III and IV present the results and discussion, respectively. Finally, conclusions are given in Section V.

II. Dataset and Method

A. Dataset

This study involves three parts of data: 1) data used for calibrating the NMCRTA; 2) data used for testing the NMCRTA; and 3) data used for validating the NMCRTA. Specifically, the NMCRTA requires on-orbit observed LST and constructed reference LSTs for practical applications. Since on-orbit real-time processing satellites are still in the experimental stage currently, there is no available on-orbit observed LST or constructed reference LST data to directly drive the NMCRTA. This study adopted 100-m Landsat 8 Collection 2 Level 2 LST data as a substitute for on-orbit observed LST data and for constructing reference LST data, which were retrieved via the single-channel algorithm version 1.3.0 [35]. The LST product has been evaluated on various surfaces, achieving an accuracy of approximately 0.5–1 K [36]. It is noted that the clouds and cloud shadows in the Landsat 8 LST data have been masked out [37], [38].

On the one hand, two long-strip images from China and the United States spanning 2020–2021 (considering the latitudinal pattern of LST spatiotemporal variation) were utilized to calibrate the NMCRTA, which comprises 727 images and 496 images, as shown in Table I. On the other hand, various thermal anomaly events from different regions around the world were used to test the effectiveness of the NMCRTA, including 1449 images spanning the years 2013–2022 (all Landsat 8 LST data were downloaded from the Google Earth Engine, https://earthengine.google.com/). Furthermore, two additional auxiliary datasets were utilized to validate the accuracy of the NMCRTA: Landsat 8 true-color images and high-resolution true-color images from Google Earth (HR images, with a resolution of 0.3–2.5 m) were employed for visual interpretation. In addition, the ESA land cover climate change initiative (LC-CCI) product (resolution: 300 m) was collected to confirm the land cover types associated with the thermal anomalies detected by the NMCRTA (https://www.esa-landcover-cci.org).

B. Method

When the observed LST deviates from its normal distribution for some reason, it is referred to as a thermal anomaly, which can be detected by analyzing the difference between the observed LST and the reference value. Existing methods focus primarily on determining thresholds, but they employ a simple mathematical averaging of adjacent spatial or temporal dimensions to establish the reference LST [39]. For the detection of low-temperature thermal anomalies, constructing a more physical reference LST and establishing a more robust threshold are crucial. Here, a novel thermal anomaly detection method named NMCRTA is proposed. The establishment and evaluation of the NMCRTA can be divided into three steps, as illustrated in Fig. 1.

Considering that using different reference temperatures can enhance the stability of thermal anomaly detection, step 1 simultaneously constructs a fit reference temperature and a real reference temperature. The fit reference temperature is the spatiotemporal trend surface of LST (STSLST, constructed from the previous year's LST data). The real reference temperature is established by averaging the contextual LST. Step 2 involves detecting thermal anomalies based on the differences between the observed temperature and both the fit and real reference temperatures, using specific thresholds. The difference between the observed temperature and the fit reference

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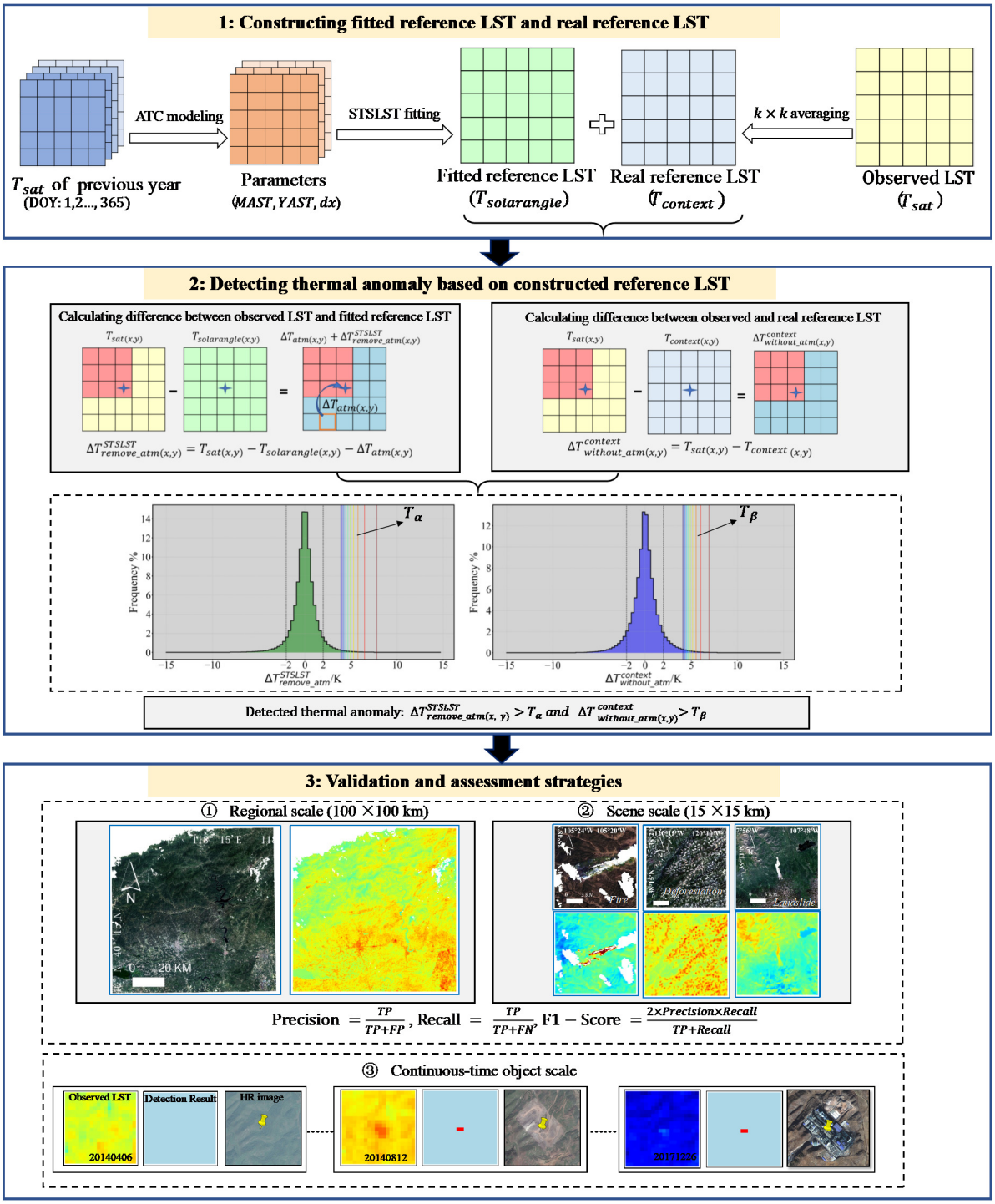


Fig. 1. Flowchart of the establishment and evaluation of NMCRTA.

temperature is corrected using the method of neighborhood median replacement of the temperature difference. The difference between the observed LST and the real reference LST can be directly calculated. The thresholds for detecting thermal anomalies were determined by two long-strip images and various typical thermal anomaly cases. Step 3 validates the effectiveness of the NMCRTA, including three strategies (regional scale, scene scale, and continuous-time object scale). A 100 × 100 km region and four 15 × 15 km scenes were used to evaluate the accuracy of this method. Additionally, two objects were used to test the reliability of the NMCRTA in detecting continuous-time variations of thermal anomalies.

1) Constructing the Fit Reference LST and Real Reference LST:

a) Fit reference LST: The widely used three-parameter ATC model can accurately and effectively fit the annual variation of the LST for a single pixel [39], [40], [41], [42], offering the advantage of easy storage in the limited storage

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TABLE I
OVERVIEW OF THE LANDSAT 8 LST IMAGES (2672 IMAGES) USED IN THIS STUDY

Image usage	Path/row	Acquisition Date	number	Location
Calibrating NMCRTA	123/023~045	01/2020-12/2021	727	Heilongjiang, Inner Mongolia, Hebei, Henan, Hubei, Hunan, and Guangdong in China
	033/026~038	01/2020-12/2021	496	North Dakota, South Dakota, Nebraska, Colorado, New Mexico in America
Testing NMCRTA	122/032	01/2013-12/2019	137	Hebei, China
	033/035	01/2021-12/2022	22	New Mexico, America
	043/033	01/2019-12/2021	66	California, America
	035/033	01/2013-12/2014	38	Colorado, America
	177/034	01/2020-12/2021	45	Mugla, Turkey
	046/029	01/2020-12/2021	35	Oregon, America
	098/085	01/2019-12/2020	42	Kangaroo Island, Australia
	226/068	01/2014-12/2015	40	Mato Grosso, Brazil
	218/076	01/2014-12/2015	41	Rio De Janeiro, Brazil
	138/217	01/2020-12/2021	36	Washington, USA
	139/208	01/2020-12/2021	55	California, America
	141/41	01/2014-12/2015	42	Bagmati, Nepal
	046/031	01/2019-12/2020	39	Oregon, America
	045/033	01/2016-12/2017	45	California, America
	183/034	01/2020-12/2021	43	Athens, Greece
	202/032	01/2020-12/2021	43	Navais Curiar, Spain
	047/024	01/2020-12/2021	44	British Columbia, Canada
	043/033	01/2020-12/2021	45	California, America
	046/016	01/2022-12/2023	41	Northwest Territories, Canada
	136/041	01/2016-12/2020	90	Assam, India
	045/030	01/2016-12/2017	42	Oregon, America
	044/021	01/2016-12/2017	45	Alberta, Canada
	021/049	01/2016-12/2017	46	Chiapas, Mexico
	125/050	01/2017-12/2018	36	Dak Lak, Vietnam
	047/027	01/2018-12/2019	36	Washington, America
	044/033	01/2018-12/2019	44	California, America
	138/041	01/2019-12/2020	38	Punakha, Bhutan
	137/041	01/2014-12/2016	63	Trongsa, Bhutan
	139/041	01/2015-12/2016	42	Sagarmatha, Nepal
	135/040	01/2015-12/2016	68	Xizang, China
		01/2022-12/2023		

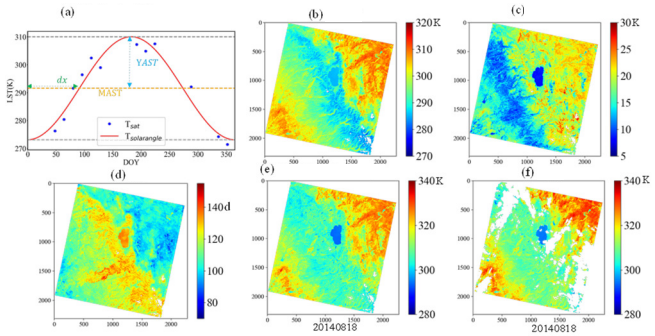


Fig. 2. Example of calculating STSLST. (a) Single pixel $T_{\text{solarangle}}$. (b) MAST. (c) YAST. (d) dx. (e) STSLST $T_{\text{solarangle}}$. (f) T_{sat}.

space of on-orbit satellites. The database of global 100-m ATC parameters needs only approximately 230 GB of storage space, representing a 120-fold compression ratio compared to daily LST data storage requirements. The equation of the ATC model is as follows:

$$T_{\text{solarangle}}(\text{DOY}) = \text{MAST} + \text{YAST} \times \sin\left(2\pi \times \frac{\text{DOY}}{365} + \text{dx}\right) \tag{1}$$

where $T_{\text{solarangle}}$ represents the fit LST, DOY represents the DOY, and the parametersMAST, YAST, and dx represent the mean, amplitude, and phase shift of the annual LST, respectively, as shown in Fig. 2(a). The three parameters are determined by fitting all the observed LST data from the entire year. Once the parameters are determined, the fit LST for a single pixel can be obtained using the ATC model, which can reflect the temporal variation trend of land surface thermal property [43], [44], [45].

For a given day, all fit LSTs within a specific spatial range collectively form a temperature trend surface that can characterize the spatial heterogeneity of the LST across different land properties. This STSLST can reflect the climatological LSTs of observed pixels, which serves as a fit reference for thermal anomaly detection. It is calculated as follows: 1) all the Landsat 8 LST images are collected for a single tile throughout the year; 2) the LST of all the pixels in the tile is fit to obtain a parameter image that includes three bands (MAST, YAST, and dx); and 3) the STSLST is obtained using the ATC model and the three-band image. Taking the tile path/row: 043/033 as an example, the process of calculating STSLST is shown in Fig. 2(b)–(e). The fit image [$T_{\text{solarangle}}$, see Fig. 2(e)] closely resembles the observed image [T_{sat}, see Fig. 2(f)] visually.

b) Real reference LST: On the basis of Tobler's first law of geography (i.e., the LST of a central location is more closely related to the nearby landscape [46]), the construction of a second reference LST is presented in (2). Although the LST of an observed pixel may be either normal or anomalous, the mean LST of the surrounding pixels within a certain neighborhood is typically normal [47] and can serve as the reference LST for the pixel. Compared with the fit STSLST, the average LST of the surrounding pixels is referred to as the

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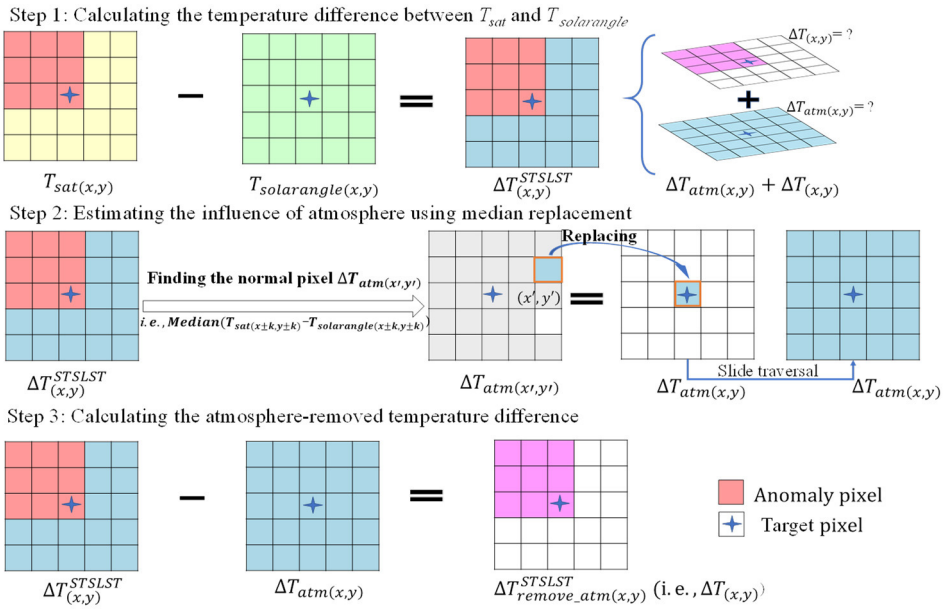


Fig. 3. Three steps for the estimation of the atmosphere-removed temperature difference.

real reference LST

$$T_{\text{context}_{(x,y)}} = \frac{\sum_{i\neq x, j\neq y} T_{\text{sat}(i,j)}}{k^2 - 1} \tag{2}$$

where (x, y) represents the location of the observed pixel; T_{context} represents the real reference LST; T_{sat} represents the observed LST obtained from satellites; and k represents the size of the context window ($k \times k$).

2) Detecting Thermal Anomalies Based on Constructed Reference LSTs:

a) Calculating the difference between the observed LST and fit reference LST: The observed LST from satellites ($T_{\text{sat}(x,y)}$) can be considered the sum of three components: the LST term determined by the STSLST from the previous year (The STSLST can reflect the climatological LST as a fit reference temperature; In real-time remote sensing detection, it is impossible to obtain the ATC-fit LST from the current year.), the LST term influenced by atmospheric conditions, and the LST term caused by a land surface thermal anomaly, as shown in the following equation:

$$T_{\text{sat}(x,y)} = T_{\text{solarangle}(x,y)} + \Delta T_{\text{atm}(x,y)} + \Delta T_{(x,y)} \tag{3}$$

where $T_{\text{solarangle}(x,y)}$ represents the STSLST; $\Delta T_{\text{atm}(x,y)}$ represents the influence of atmospheric conditions (e.g., weather), and $\Delta T_{(x,y)}$ represents the LST variation that cannot be explained by the STSLST and atmospheric conditions (e.g., thermal anomaly events).

Fig. 3 illustrates the calculation of the temperature difference between the observed LST and the fit reference LST (i.e., step 1, $T_{\text{sat}(x,y)}$-$T_{\text{solarangle}(x,y)}$), the estimation of $\Delta T_{\text{atm}(x,y)}$ (i.e., step 2), and the calculation of the remainder term$\Delta T_{(x,y)}$ for thermal anomaly detection (i.e., step 3).

As shown in (4) (i.e., step 1 in Fig. 3), the directly calculated temperature difference ($\Delta T^{\text{STSLST}}_{(x,y)}$) comprises two items: $\Delta T_{\text{atm}(x,y)}$ and $\Delta T_{(x,y)}$. It is difficult to remove the $\Delta T_{\text{atm}(x,y)}$ term from $\Delta T^{\text{STSLST}}_{(x,y)}$ to obtain a temperature difference without the influence of atmospheric conditions

$$\Delta T^{\text{STSLST}}_{(x,y)} = T_{\text{sat}(x,y)} - T_{\text{solarangle}(x,y)} = \Delta T_{\text{atm}(x,y)} + \Delta T_{(x,y.)}. \tag{4}$$

For that, a new algorithm based on the neighborhood median replacement of the temperature difference is proposed to estimate the $\Delta T_{\text{atm}(x,y)}$ item (see step 2 in Fig. 3). For a pixel (x, y) experiencing a thermal anomaly, its $\Delta T_{\text{atm}(x,y)}$ term is equivalent to that of a normal pixel (x', y') within its neighborhood, but it is difficult to determine the location of (x', y'). For the normal pixel (x', y'), $\Delta T_{\text{atm}(x',y')}$ can be calculated by $T_{\text{sat}(x',y')} - T_{\text{solarangle}(x',y')}$. Here, we suppose that the location has a temperature difference equal to the median value of all temperature differences within the neighborhood [i.e., Median$(T_{\text{sat}(x\pm k,y\pm k)} - T_{\text{solarangle}(x\pm k,y\pm k)})$] could be regarded as the (x', y'), when more than half of the neighboring pixels of (x, y) are normal. Therefore, the $\Delta T_{\text{atm}(x,y)}$ of the pixel experiencing a thermal anomaly can be estimated by the following equation:

$$\begin{aligned}\Delta T_{\text{atm}(x,y)} &= \Delta T_{\text{atm}(x',y')} \\ &= \text{Median}\left(T_{\text{sat}(x\pm k,y\pm k)} - T_{\text{solarangle}(x\pm k,y\pm k)}\right).\end{aligned} \tag{5}$$

This new algorithm can also be applied to pixel (x, y) experiencing a normal state to estimate its $\Delta T_{\text{atm}(x,y)}$. k represents the size of the neighborhood (it was set to 15 to achieve a balance between accuracy and computing time). The ΔT_{atm} of each pixel within the neighborhood can be obtained through slide traversal using this algorithm. Then, the $\Delta T_{\text{atm}(x,y)}$ item could be removed from $\Delta T^{\text{STSLST}}_{(x,y)}$, as shown in (6) (i.e., step 3 in Fig. 3). The atmosphere-removed temperature difference ($\Delta T^{\text{STSLST}}_{\text{remove_atm}(x,y)}$, i.e.,$\Delta T_{(x,y)}$) is finally obtained

$$\begin{aligned}&\Delta T^{\text{STSLST}}_{\text{remove_atm}(x,y)} \\ &= \Delta T^{\text{STSLST}}_{(x,y)} - \Delta T_{\text{atm}(x,y)} \\ &= \Delta T^{\text{STSLST}}_{(x,y)} - \text{Median}\left(T_{\text{sat}(x\pm k,y\pm k)} - T_{\text{solarangle}(x\pm k,y\pm k)}\right).\end{aligned} \tag{6}$$

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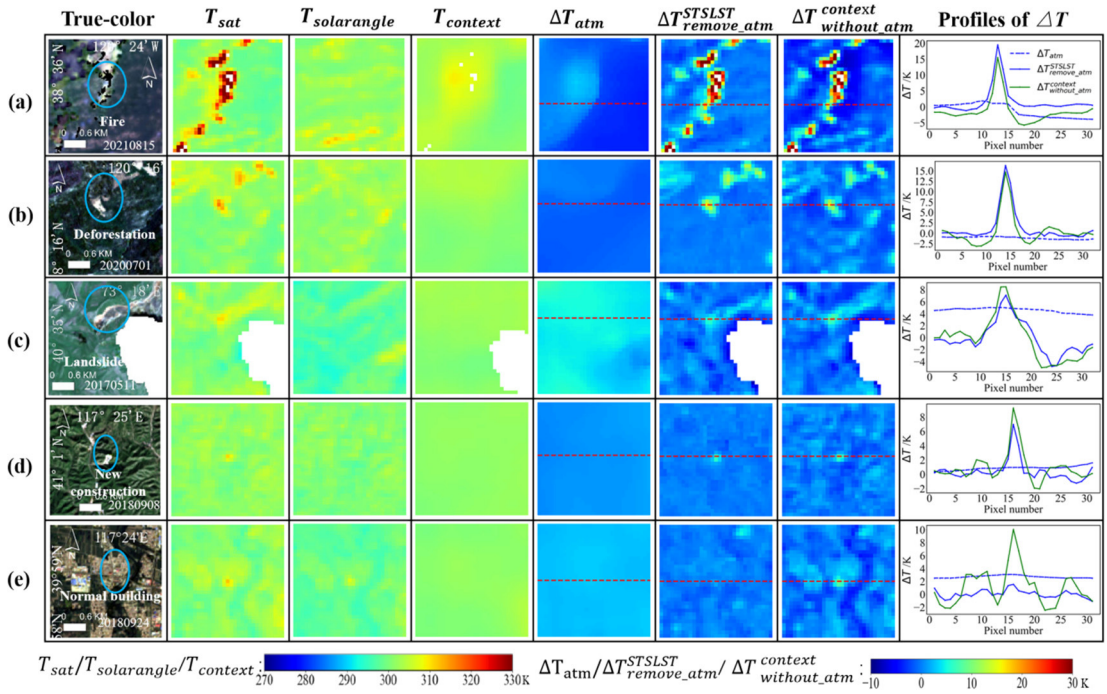


Fig. 4. Examples of four thermal anomaly events and one normal case. The profiles of ΔT in the last column represent the values along the red line. (a) Fire scenario. (b) Deforestation scenario. (c) Landslide scenario. (d) New construction scenario. (e) Normal building scenario.

b) Calculating the difference between the observed and real reference LST: Different from STSLST, similar atmospheric conditions exist between the observed LST of the target pixel and the average LST of the contextual pixels, as they are obtained simultaneously. The temperature difference between the observed LST and the real reference LST (i.e., $\Delta T^{\text{context}}_{\text{without_atm}(x,y)}$) can be directly obtained via the following equation:

$$\Delta T^{\text{context}}_{\text{without_atm}(x,y)} = T_{\text{sat}(x,y)} - \frac{\sum_{i\neq x, j\neq y} T_{\text{sat}(i,j)}}{k^2 - 1}. \tag{7}$$

c) Detecting thermal anomalies through temperature differences and corresponding thresholds: For thermal anomaly events, the observed temperature usually exhibits varying degrees of difference compared to the reference temperature. As shown in Fig. 4, for high-temperature fire thermal anomalies [Fig. 4(a)], both the $\Delta T^{\text{STSLST}}_{\text{remove_atm}}$ and $\Delta T^{\text{context}}_{\text{without_atm}}$ values significantly increase, with the temperature differences exceeding 15 K. Furthermore, for low-temperature thermal anomalies [Fig.4(b)–(d)] such as deforestation, landslides, and newly constructed buildings, both the $\Delta T^{\text{STSLST}}_{\text{remove_atm}}$ and $\Delta T^{\text{context}}_{\text{without_atm}}$ exceed 6 K. These thermal anomaly events exhibit a significant increase in both $\Delta T^{\text{STSLST}}_{\text{remove_atm}}$ and $\Delta T^{\text{context}}_{\text{without_atm}}$ simultaneously. For normal buildings suspected of thermal anomalies [Fig. 4(e)], there is a significant increase in $\Delta T^{\text{context}}_{\text{without_atm}}$[exceeding 6 K)] while $\Delta T^{\text{STSLST}}_{\text{remove_atm}}$ shows no significant increase (not exceeding 2 K). The thermal anomaly detection results can be obtained by comparing the two temperature differences with their respective thresholds T_α and T_β(T_α represents the threshold of $\Delta T^{\text{STSLST}}_{\text{remove_atm}}$, $T_\beta$represents the threshold of $\Delta T^{\text{context}}_{\text{without_atm}}$). These thresholds are determined based on two long-strip images and various typical thermal anomaly cases. On the basis of the principle that anomalies typically follow a long-tailed distribution [48], T_α and T_β were set to 5.76 and 5.52 K, respectively, in this study (the details of threshold determination are introduced in Section IV-B).

3) Validation and Assessment Strategies: To comprehensively assess the effectiveness of the NMCRTA, this study conducted experiments at the single-temporal regional scale, single-temporal scene scale, and continuous-time object scale. At the regional scale, a 100 × 100 km sample region of Tangshan, Hebei, China (central coordinates: 118.07E, 40.378N) with diverse land covers was selected to assess the accuracy of this method. At the scene scale, four 15 × 15 km thermal anomaly scenarios (fire, deforestation, landslide, and industrial heat source) were used to test the effectiveness of this method. Specifically, 5000/1000 sample points were randomly and evenly selected at the region/scene scale. The detection results of these sample points were subsequently confirmed with the help of auxiliary data. Finally, the precision, recall, and $F1$-Score were calculated to evaluate the accuracy of the NMCRTA as follows [49]:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{8}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{9}$$

$$F1 - \text{Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{10}$$

where TP represents the number of points correctly identified as thermal anomalies, FP represents the number of points incorrectly identified as thermal anomalies, and FN represents the number of points incorrectly identified as nonthermal anomalies. At the continuous-time object scale, two objects

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were used to evaluate the reliability of the NMCRTA in capturing the continuous-time variations in thermal anomaly events from 2014 to 2017. For a continuous-time object, the $F1$-Score cannot be calculated. Here, the continuous detection of thermal anomalies for a target object was evaluated through high-resolution (HR) images and related auxiliary materials (i.e., HR and auxiliary documentation).

During the evaluation, the widely used visual interpretation method was employed to confirm the detection results [31], [50]. Specifically, this study used visual interpretation of current and previous year images to determine surface anomaly events. Because the surface anomaly events obtained from visual interpretation of optical images are not exactly equivalent to the thermal anomaly events detected, the following four rules were proposed.

1) The detected result is deemed correctly identified when a natural disaster event (e.g., fires or landslides) and a thermal anomaly occur simultaneously.

2) The detected result is deemed correctly identified when an ecological damage event (e.g., deforestation or new building construction) and a thermal anomaly happen simultaneously.

3) The detected result is not deemed a thermal anomaly event when it relates to agricultural activity (e.g., sowing or harvesting). A masking process is applied.

4) The detected result is deemed misidentified (nonthermal anomalies) when it does not belong to the three categories mentioned above.

III. Result

A. Detected Thermal Anomalies at the Regional Scale

A 100 × 100 km region of Tangshan, Hebei, China, was selected to evaluate the accuracy of NMCRTA in complex land-cover environments, including forests, grasslands, water, bare soil, and impervious surfaces. The LST image of the selected region obtained on August 12, 2014, was selected as input, considering its cloud-free coverage. Fig. 5(a) displays the thermal anomaly detection results using the NMCRTA, along with the confirmation through visual interpretation. Fig. 5(b) shows the detailed thermal anomaly detection process, including the observed temperature, two constructed reference temperatures, and detection results. Correctly and incorrectly identified thermal anomalies, indicated by red and blue-green dots in the figure, respectively, were determined through visual interpretation of true-color Landsat 8 and HR images. For ease of display, the detection results of the same thermal anomaly event are presented together on the right side of Fig. 5.

Fig. 5(a) shows that NMCRTA identified 168 thermal anomaly points, of which 151 were correctly identified and 17 were incorrectly identified, resulting in an accuracy of 89.88% in detected samples. The 151 confirmed thermal anomaly points were distributed across 39 thermal anomaly events, including 28 caused by deforestation, six caused by newly constructed buildings, and five caused by river drying. They are typical low-temperature thermal anomaly events, compared to widely studied high-temperature thermal anomaly events, such as fires and volcanic eruptions. For example, these thermal anomaly points ID1-7 of Fig. 5(a) in 2013 were covered with forest, whereas they appeared as deforestation areas on August 12, 2014. It can be explained by Fig. 5(b) that deforestation led to a noticeable increase in the observed T_{sat} compared with the reference temperatures $T_{\text{solarangle}}$ and T_{context}. Similarly, the thermal anomaly points ID 48-51 in Fig. 5(a) were bare land in 2013, whereas a new building was observed in 2014. It can be found that a significant increase in the observed temperature compared with the reference temperatures [Fig. 5(b)]. Furthermore, points ID 84-121 in Fig. 5(a) represented the river water body in 2013; however, river drying in 2014 caused the riverbed to become exposed. The exposed riverbed led to a significant increase in the observed LST compared with reference temperatures, as shown in Fig. 5(b). These anomaly events can lead to a noticeable increase in the observed T_{sat} from the above cases, which could be detected by NMCRTA. For the rare misidentified thermal anomaly points [see blue-green points in Fig. 5(a)], the land cover was mainly bare land. The possible reason for this misidentification is the decreased moisture content of the bare land, which leads to a reduction in heat capacity and an increase in observed LST. Nevertheless, this condition is relatively rare.

To further comprehensively evaluate the effectiveness of NMCRTA in the entire test region, 5000 detected sample pixels were randomly selected in the 100 × 100 km region. The calculated precision, recall, and $F1$-Score were 86.87%, 81.25%, and 0.838, respectively. It is worth noting that some unidentified thermal anomaly samples are due to the difference in spatial resolution between LST images and true-color images. These thermal anomalies are small in spatial range, which can be observed in true-color images with a spatial resolution of 30 m, but are undetectable in LST images with a 100-m resolution. Overall, the NMCRTA demonstrated reliable accuracy in detecting thermal anomalies within the complex land-cover environments at a 100 × 100 km regional scale.

B. Detected Thermal Anomalies at the Scene Scale

Four types of thermal anomaly scenario data were derived from forest fire imagery of New Mexico on April 12, 2022, deforestation imagery of California on July 1, 2021, landslide imagery of Colorado on May 25, 2014, and industrial heat source imagery of Hebei on July 11, 2014. All scenarios covered a spatial range of 15 × 15 km with cloud-masked images. Fig. 6 shows the detection results for different thermal anomaly scenarios via the NMCRTA. The red box in the figure represents the enlarged detailed area. Visual evaluation was conducted on the overall region and the detailed area, taking the true-color and observed LST images as references. Furthermore, 1000 sample points were randomly and evenly selected from the first three thermal anomaly scenarios to quantitatively evaluate the accuracy of NMCRTA (as detailed in Section II-C3), respectively.

In the visual evaluation, the active fire area exhibited a southwest-to-northeast trend, with some scattered active fire spots present in the northwest [Fig. 6(a1)–(a5)]. The spatial distribution pattern of the detection results [Fig. 6(a6)] was consistent with the visual interpretation. In the deforestation case shown in Fig. 6(b1)–(b6), visual interpretation revealed new deforested areas in the central region and northeast [Fig. 6(b1)] compared with the reference image

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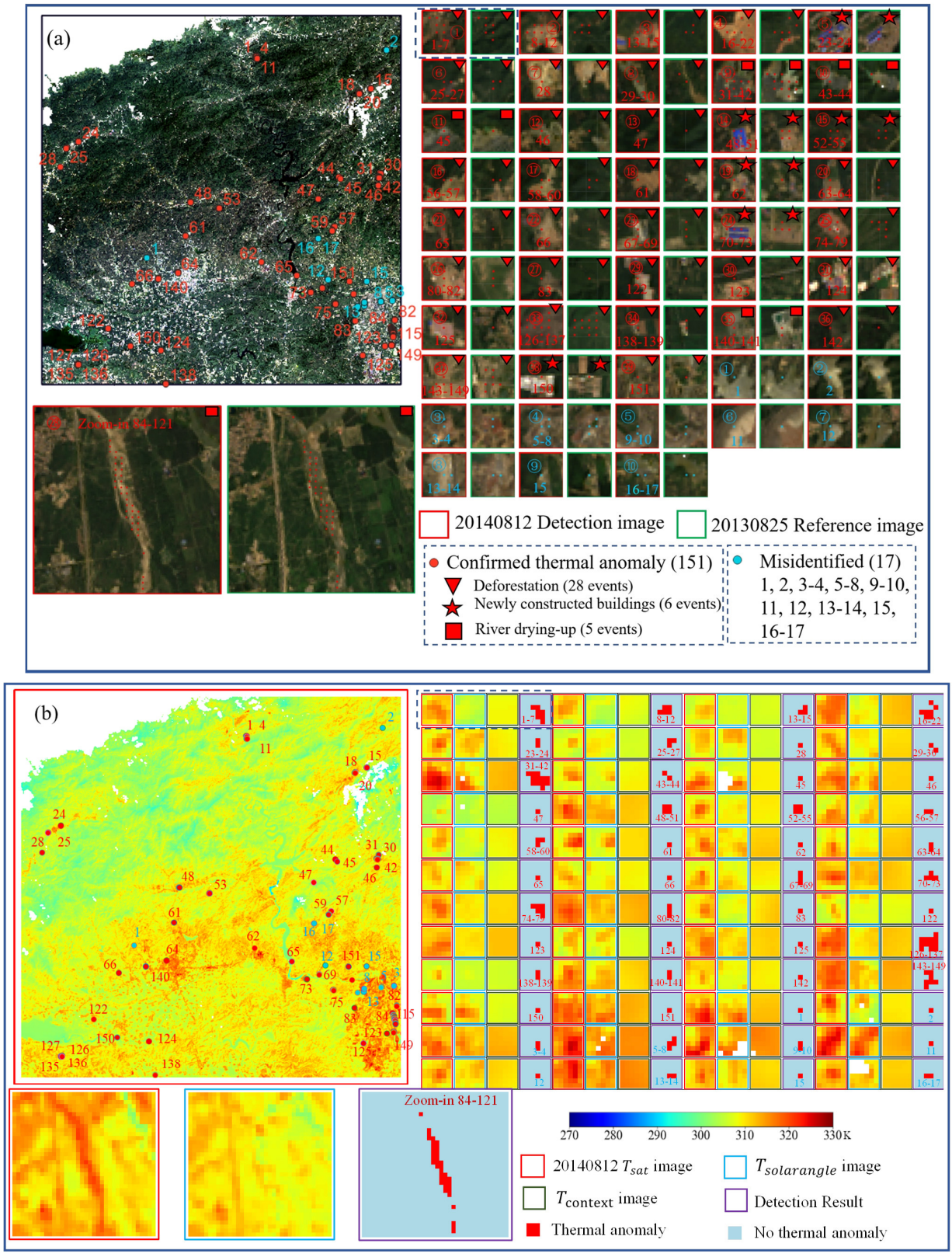


Fig. 5. Results of thermal anomaly detection at the 100 × 100 km regional scale in Tangshan (central coordinates: 118.07E, 40.378N). (a) Confirmation of the detection results. The true-color images were obtained on August 25, 2013, and August 12, 2014. Specifically, the 28 deforestation events correspond to the following thermal anomaly points: ① 1–7, ② 8–12, ③ 13–15, ④16–22, ⑥ 25–27, ⑦ 28, ⑧ 29–30, ⑫ 46, ⑬ 47, ⑯ 56–57, ⑰ 58–60, ⑱ 61, ⑳ 63–64, ㉑65, ㉒66, ㉓67–69, ㉕74–79, ㉖80–82, ㉗83, ㉙122, ㉚123, ㉛124, ㉜125, ㉝126–137, ㉞138–139, ㊱142, ㊲143–149, and ㊴151. The 6 newly constructed buildings correspond to the following anomaly points: ⑤ 23–24, ⑭ 48–51, ⑮ 52–55 ⑲ 62, ㉔70–73, and ㊳150. The 5 river drying events are associated with the following points: ⑨ 31–42, ⑩ 43–44, ⑪ 45, ㉘84–121, and ㉟140–141. (b) Display of the detection results.

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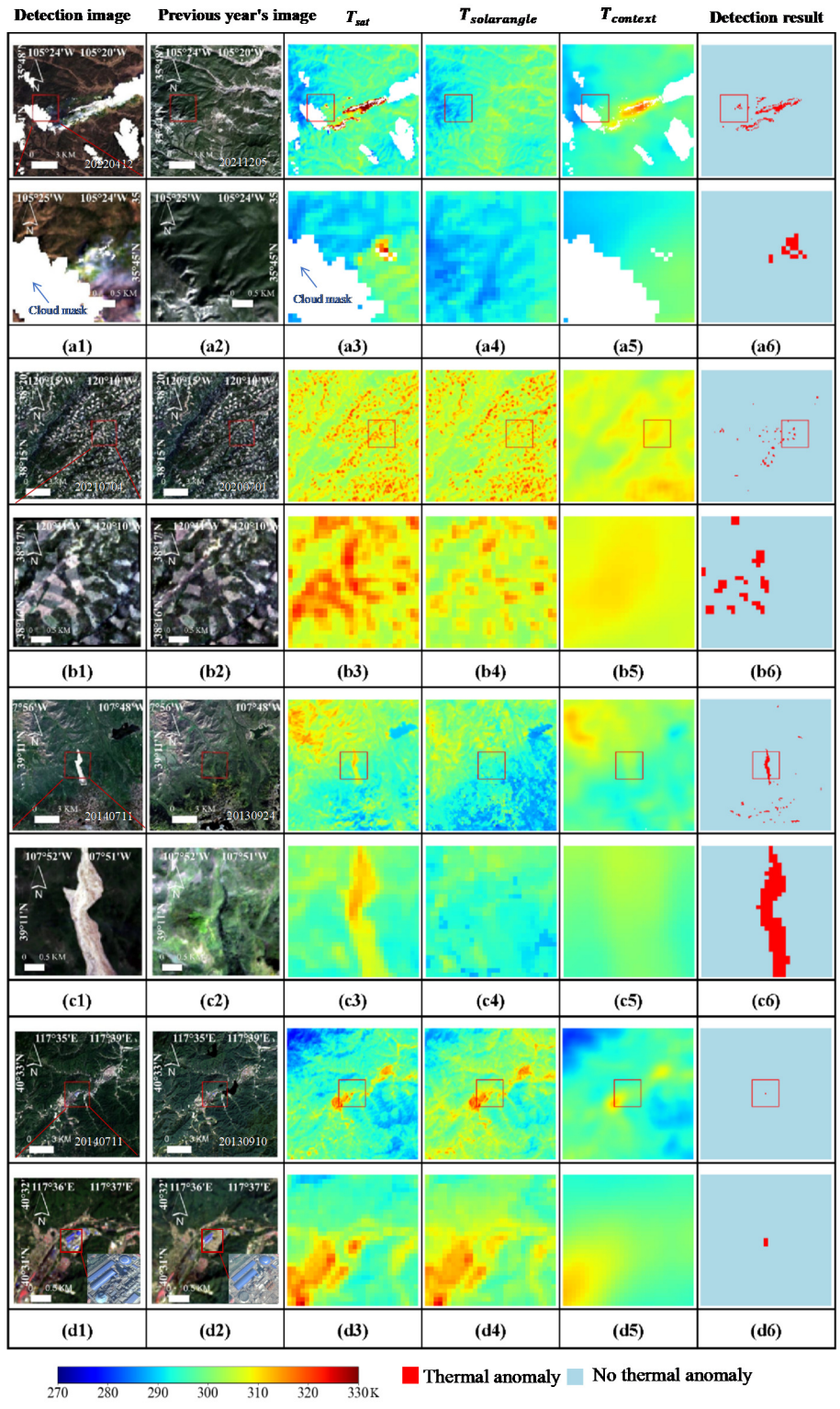


Fig. 6. Detection of different thermal anomaly scenarios via the NMCRTA. The first and second columns represent the observed image and the reference image from the previous year, respectively. The third column represents the T_{sat} image. The fourth and fifth columns represent the $T_{\text{solarangle}}$ image and the T_{context} image, respectively. The sixth column represents the thermal anomaly detection results. (a1)–(a6) Fire scenario. (b1)–(b6) Deforestation scenario. (c1)–(c6) Landslide scenario. (d1)–(d6) Industrial heat source scenario.

[Fig. 6(b2)]. The detection results [Fig. 6(b6)] indicate that NMCRTA can effectively identify recently deforested areas without misidentifying previously deforested areas (from the previous year) as thermal anomalies. Similarly, a comparison between the detection image and the reference image revealed a landslide in the central region and sporadic deforestation in the southern region, as illustrated in Fig. 6(c1)–(c5). The detected landslide area [Fig. 6(c6)] in the central region

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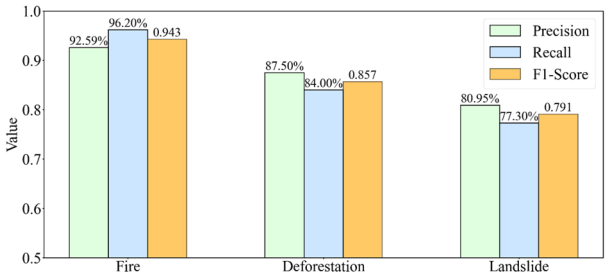


Fig. 7. Precision, recall, and $F1$-Score of the fire deforestation and landslide anomaly scenarios.

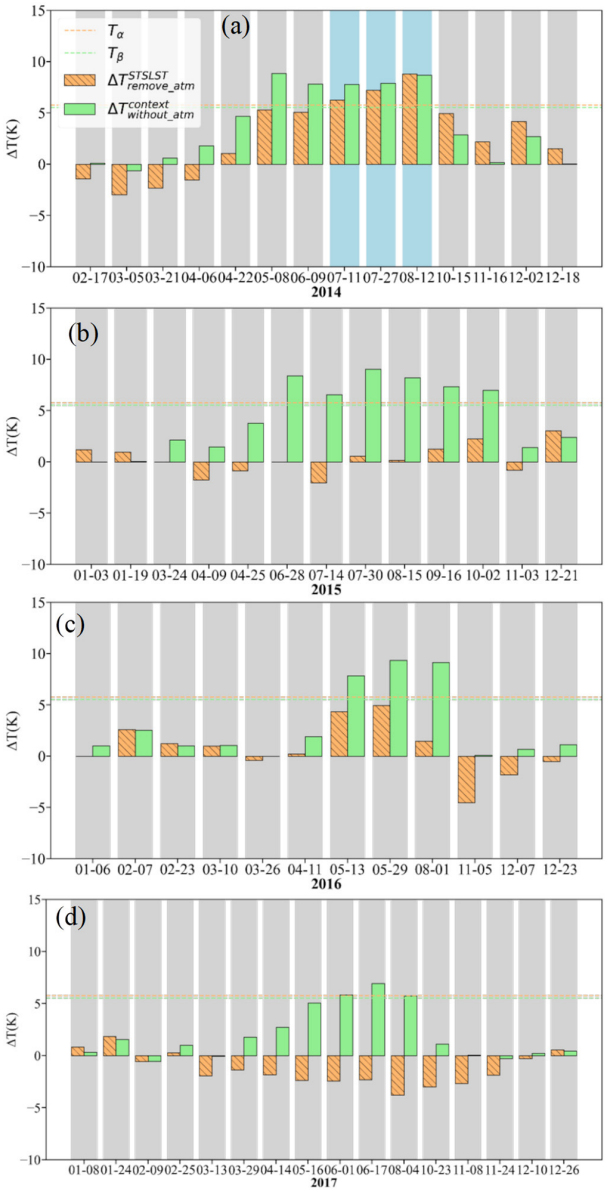


Fig. 8. Continuous-time detection results of thermal anomalies associated with newly constructed buildings. (a)–(d) 2014–2017 temperature differences between the observed LST and two reference LSTs. The light blue background indicates the detection of a thermal anomaly.

corresponds accurately with the actual region. Fig. 6(d1)–(d6) shows that a newly constructed cement plant was detected, while residential areas were not identified. Auxiliary material

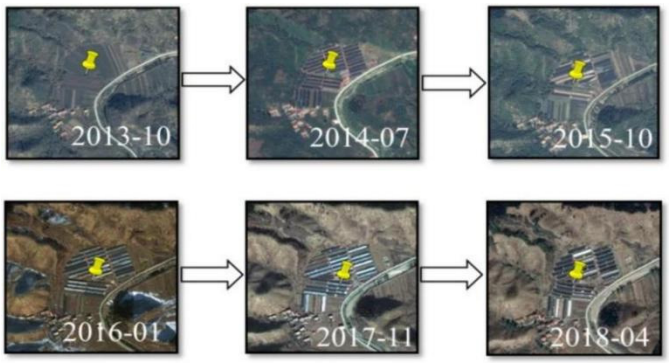


Fig. 9. HR images of the construction of new buildings.

indicates that this cement plant was under construction in 2013 and began operating as an industrial heat source in 2014 (http://www.cbminfo.com/BMI/xyljqyfzdt/jddt/6510415/index.html). NMCRTA is effective in detecting industrial heat sources.

Fig. 7 presents the quantitative evaluation results, including the precision, recall, and $F1$-Score. The industrial heat source scenario was only evaluated through visual interpretation due to the scarcity of thermal anomaly samples [Fig. 6(d6)]. In the quantitative evaluation, fire detection achieved the highest accuracy, with both the precision and recall exceeding 90% and an $F1$-Score of 0.943 (Fig. 7). Deforestation detection achieved secondary accuracy, with both the precision and recall exceeding 84%, which resulted in an $F1$-Score of 0.857. Most of the newly deforested areas were successfully detected by NMCRTA. For the landslide detection, the precision, recall, and $F1$-Score were 80.95%, 77.30%, and 0.791, respectively. From a physical perspective, the thermal anomaly observed in fires can be attributed to intense radiative energy emission, while those associated with deforestation and landslides primarily result from the land surface cover alterations. It can be observed that NMCRTA effectively detects areas with a significant increase in LST after a landslide event. Overall, although the $F1$-Score of deforestation and landslide detection may not be as high as that of fire detection, their precision exceeds 80%. As a supplement to optical detection of land surface anomalies, this study employs thermal infrared data to detect different degrees of thermal anomalies (such as fire, deforestation, landslides, and industrial heat sources), which offers a reliable way for real-time remote sensing detection of these anomalies both day and night.

C. Detected Thermal Anomalies in the Continuous-Time Object Scale

To assess the sensitivity of the NMCRTA in detecting continuous-time variations of thermal anomalies, two thermal anomaly objects from the Tangshan sample region (ID 52 and ID 74 in Fig. 5) were selected. The two thermal anomaly objects are newly constructed buildings and deforestation, with center coordinates of 118.064°E, 40.359°N and 118.443°E, 40.127°N, respectively.

Figs. 8 and 9 show the continuous-time detection results (cloud-masked images from 2014 to 2017) and HR images

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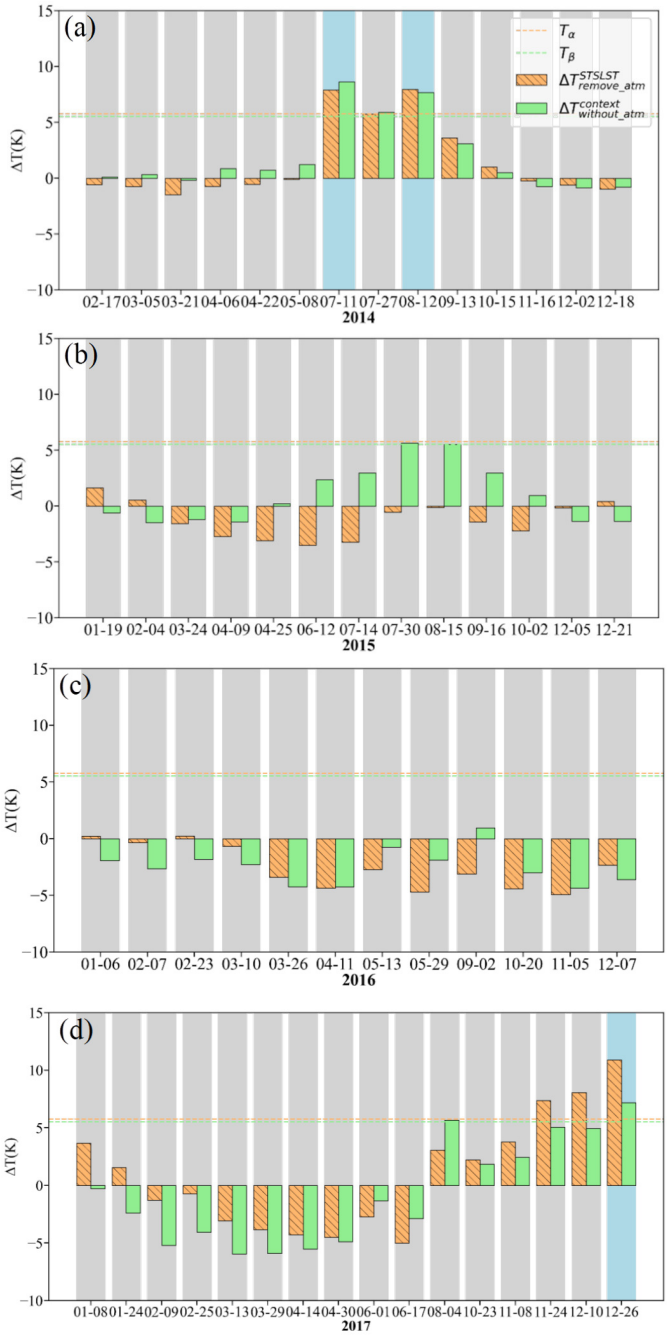


Fig. 10. Continuous-time detection results for thermal anomalies caused by deforestation. (a)–(d) 2014–2017 temperature differences between the observed LST and two reference LSTs. The light blue background indicates the detection of a thermal anomaly.

of newly constructed buildings. As shown in Fig. 8(a)–(d), thermal anomalies were detected on July 11 and 27, 2014, and August 12, 2014, while no thermal anomaly was detected in 2015, 2016, or 2017 for the newly constructed buildings. Some candidate thermal anomalies were repeatedly detected via the real reference temperature from 2015 to 2017, whereas none were identified via the STSLST reference temperature [Fig. 8(b)–(d)]. This discrepancy can be attributed to changes in land properties (from low-temperature cropland to high-temperature buildings) and annual updates of the STSLST

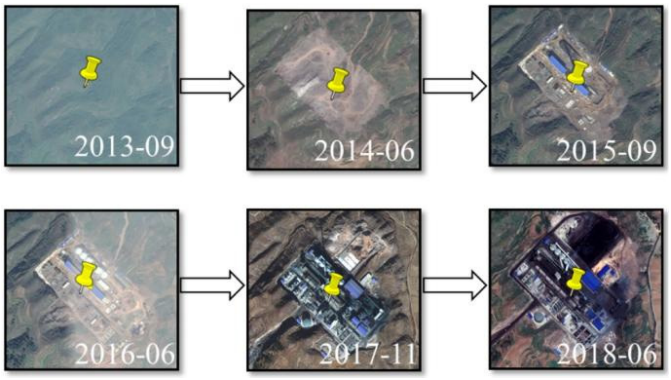


Fig. 11. HR images showing the transitions from forest to bare land and then to a factory (coking plant).

(which represents the thermal characteristics of the newly constructed buildings). The HR images indicated that this area was cropland in 2013, newly constructed buildings in 2014, and existing buildings (no industrial heat source factory) in 2015, 2016, and 2017 (Fig. 9). Overall, the proposed NMCRTA can effectively capture the thermal anomalies related only to buildings under construction.

Figs. 10 and 11 show the continuous-time detection results (cloud-masked images from 2014 to 2017) and HR images of deforestation. Thermal anomalies were detected on July 11, 2014, and August 12, 2014 in Fig. 10. Fig. 11 indicates that the object area was forestland in 2013, but became bare land in 2014. Additionally, sporadic factory facilities appeared at the site from 2015 to 2016. No thermal anomalies were detected in Fig. 10 (b) and (c) since these small-area factory facilities (Fig. 11) could not cause a significant increase in the LST. However, a thermal anomaly was detected again on December 26, 2017, as shown in Fig. 10(d). Although no thermal anomaly event was found in the HR images of 2017 (Fig. 11), an administrative document reveals that a factory (coking plant) began operating in October 2017 and then manifested as a new industrial heat source(http://xzspj.tangshan.gov.cn/index.php?m=content&c=index&a=show&catid=29&id=4038). In summary, the proposed NMCRTA can capture different thermal anomalies associated with a factory.

IV. Discussion

A. Reliability of the STSLST in Characterizing the Spatial Heterogeneity of LST

The reliability of the constructed STSLST ($T_{\text{solarangle}}$) in characterizing the spatial heterogeneity of the previous year's LST (T'_{sat}) determines the effectiveness of the NMCRTA. When the spatial heterogeneity of the constructed STSLST aligns well with that of T'_{sat}, the LST difference ($T_{\text{solarangle}} - T'_{\text{sat}}$) becomes spatially uniform, which leads to its STD approaching zero. Thus, a smaller STD usually implies a more reliable STSLST. Taking randomly selected water (117.536 E, 40.036 N), forests (118.276 E, 40.737 N), impervious surfaces (117.952 E, 40.202 N), and mixed land cover (117.956 E, 40.292 N) in Tangshan as examples, T'_{sat}, $T_{\text{solarangle}}$, and $T_{\text{solarangle}} - T'_{\text{sat}}$ (including the average values and STD) within a spatial range of 1.5 × 1.5 km in four seasons of 2015 are given

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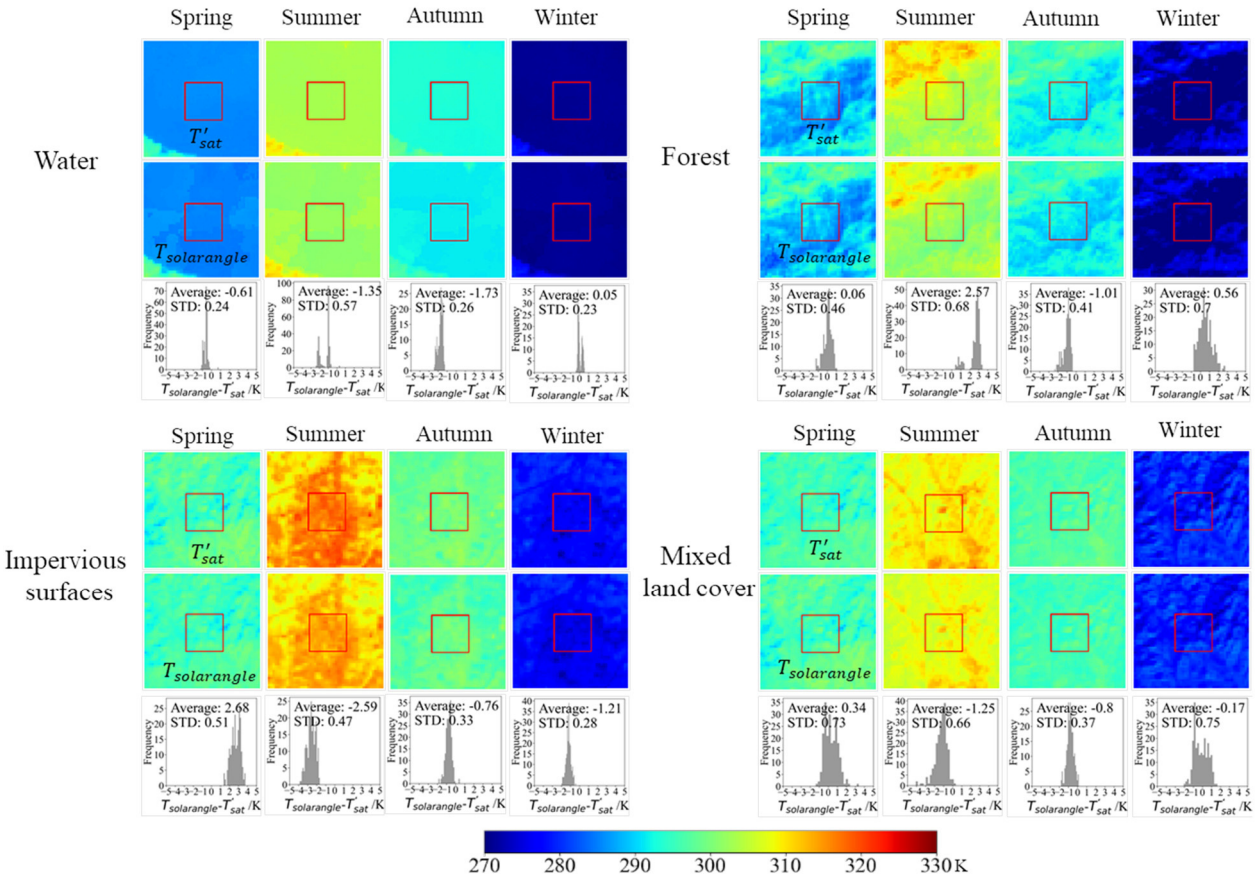


Fig. 12. Spatial distributions of T'_{sat} and$T_{solarangle}$ and for the four samples (water, forest, impervious surfaces, and mixed land cover). Spring: image from April 9, 2015; Summer: image from August 15, 2015; Autumn: image from October 2, 2015; Winter: image from January 19, 2015; The red box represents the spatial range of the sample with an area of 1.5 × 1.5 km; The histogram shows the average and STD of $T_{\text{solarangle}} - T'_{\text{sat}}$.

in Fig. 12. The spatial texture of $T_{\text{solarangle}}$ is visually similar to that of T'_{sat} across all four seasons. The average values of $T_{\text{solarangle}} - T'_{\text{sat}}$ ranged from –2.59 to 2.68 K across the four seasons, which indicates that the $T_{\text{solarangle}}$ is comparable to T'_{sat} in the whole year. More importantly, the STD of $T_{\text{solarangle}} - T'_{\text{sat}}$ was less than 0.75 K across all seasons for different land cover types. The constructed STSLST is relatively reliable for characterizing the spatial heterogeneity of T'_{sat}, since an accuracy of approximately 1.0 K can be considered as high-precision ATC fitting [42].

B. Reliability of Thresholds T_α and T_β for Thermal Anomaly Detection

As introduced in Section II-A, two long-strip sample regions in the Northern Hemisphere were selected to determine potential thresholds (Fig. 13). One of the sample regions was in China, covering 23 Landsat 8 tiles (123/023~045), while the other was in the continental United States, including 13 Landsat 8 tiles (033/026~038). They encompass various land cover types and exhibit distinct latitudinal differences, which could represent the spatial heterogeneity of LST to some extent. A total of 1223 images acquired across different seasons of 2020 and 2021 were used to calculate the differences between the observed LST and the two reference LSTs. Specifically, all 566 images from 2020 were utilized to establish the

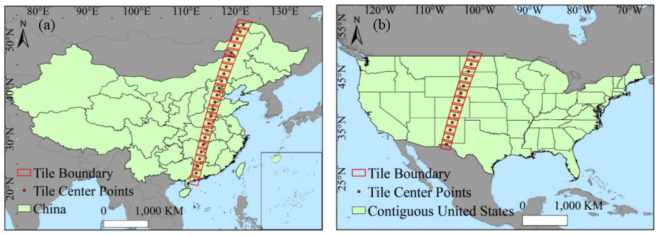


Fig. 13. Location of the sample regions. (a) China. (b) Continental United States.

STSLST, while the 657 images from 2021 were employed to calculate $\Delta T^{\text{STSLST}}_{\text{remove_atm}}$ and $\Delta T^{\text{context}}_{\text{without_atm}}$.The final threshold was determined in two steps: first, ten potential thresholds were selected based on the distribution of $\Delta T^{\text{STSLST}}_{\text{remove_atm}}$ and $\Delta T^{\text{context}}_{\text{without_atm}}$[Fig. 14(a-b)]; second, the final threshold was determined by selecting the potential threshold that yielded the highest $F1$-Score in the test dataset comprising 30 thermal anomaly scenarios across Asia, Europe, and North America.

The frequency distributions of these two temperature differences ($\Delta T^{\text{STSLST}}_{\text{remove_atm}}$ and $\Delta T^{\text{context}}_{\text{without_atm}}$) are illustrated in Fig. 14(a) and (b). It can be found that both distributions of $\Delta T^{\text{STSLST}}_{\text{remove_atm}}$ and $\Delta T^{\text{context}}_{\text{without_atm}}$ follow a normal distribution, with more than 85% of the values falling within the range of –2~2. Considering that land surface anomalies are typi-

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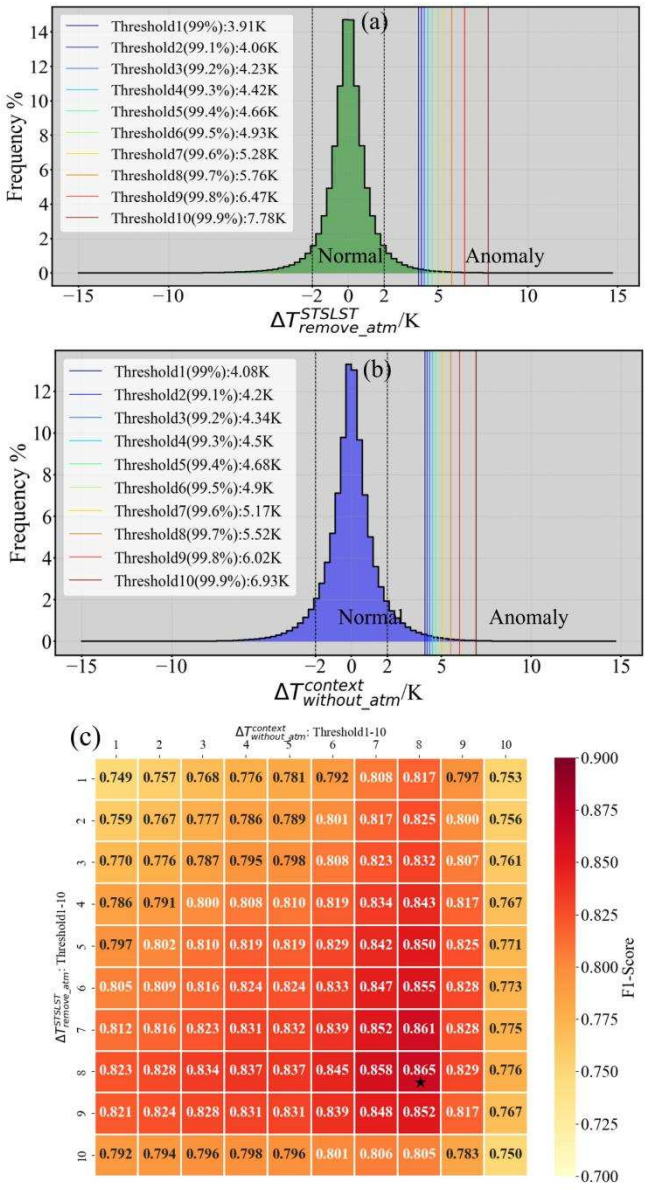


Fig. 14. Result of threshold determination. (a) Frequency distribution of $\Delta T^{\mathrm{STSLST}}_{\mathrm{remove_atm}}$. (b) Frequency distribution of $\Delta T^{\mathrm{context}}_{\mathrm{without_atm}}$. (c) Accuracy evaluation of 100 potential threshold combinations. The symbol ⋆ represents the highest $F1$-Score.

cally low-probability events [27], [48], the two temperature differences with the same cumulative frequencies ranging from 99% to 99.9% (with a step of 0.1%) were adopted as potential threshold combinations. Compared with directly determining potential thresholds using equal intervals within a predefined maximum and minimum threshold range (e.g., 1 K step within 5–10 K), the cumulative frequency-based method for determining potential thresholds in this study is more representative and more physical. Finally, ranges of 3.91–7.78 K for $\Delta T^{\mathrm{STSLST}}_{\mathrm{remove_atm}}$ and 4.08–6.93 K for $\Delta T^{\mathrm{context}}_{\mathrm{without_atm}}$ were obtained as shown in Fig. 14(a) and (b). A low threshold usually leads to over-identification, while a high threshold may lead to under-identification. The 100 potential threshold combinations (i.e., 10 $\Delta T^{\mathrm{STSLST}}_{\mathrm{remove_atm}}$ thresholds × 10 $\Delta T^{\mathrm{context}}_{\mathrm{without_atm}}$thresholds) were applied to a test dataset consisting of 30 thermal anomaly

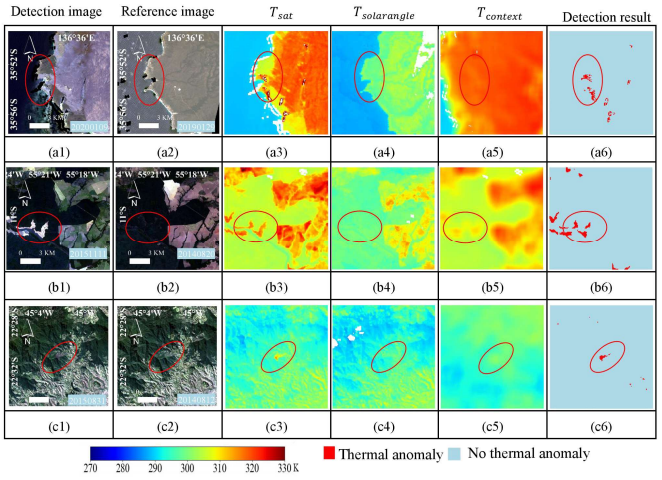


Fig. 15. Detection of various thermal anomalies in the Southern Hemisphere via the NMCRTA. (a1)–(a6) Fire detection in Kangaroo Island, Australia. (b1)–(b6) Deforestation detection in Mato Grosso, Brazil. (c1)–(c6) Landslide detection in Rio De Janeiro, Brazil.

scenarios. Each potential threshold combination could produce an $F1$-Score value. Then, the combination with the highest $F1$-Score was selected as the final optimal threshold [see the ⋆ in Fig. 14(c)]. The results show the $F1$-Score initially increases and then decreases as the threshold increases, reaching the maximum (0.865) at Threshold8 (T_α: 99.7%, T_β: 99.7%). Therefore, the Threshold8 (T_α = 5.76, T_β =5.52) was adopted for thermal anomaly detection in this study.

Moreover, the effectiveness of adopted thresholds required further validation in the Southern Hemisphere. The specific validation locations were selected based on two strategies: 1) identifying the hotspot regions for fire, deforestation, and landslide occurrences through spatial frequency analysis and 2) determining representative anomaly events through visual interpretation for the images from 2014 to 2022. Then, three representative validation regions were obtained as shown in Fig. 15 (all images were cloud-masked): Kangaroo Island (Australia) for fires, Mato Grosso (Brazil) for deforestation, and Rio de Janeiro (Brazil) for landslides. These thermal anomaly events were evaluated within a range of 15 × 15 km. In general, the detection results are consistent with the thermal anomaly region with significant temperature variation and the disaster regions in the true-color images [Fig. 15(a)–(c)]. Taking the deforestation case as an example, the area of increased LST caused by deforestation was detected after removing the influence of atmospheric conditions [Fig. 15(b3)–(b6)], which accurately corresponds to the significant change area of reflectance observed in the true-color images [highlighted by the red ellipse in Fig. 15(b1) and (b2)]. Overall, the performance of the adopted thresholds is acceptable in these typical thermal anomalies across different regions of the Southern Hemisphere.

C. Applicability of Detecting Thermal Anomalies at Nighttime

Compared to optical imaging, thermal infrared imaging has the advantage of nighttime observation. To test the effectiveness of NMCRTA in detecting nighttime thermal anomalies, this method was evaluated by two nighttime

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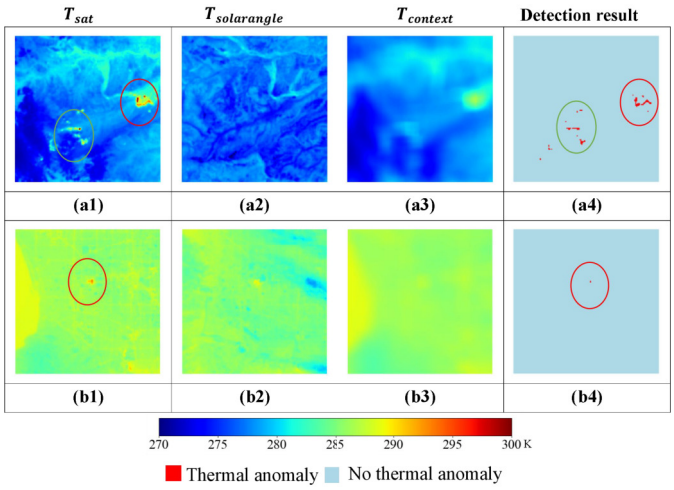


Fig. 16. NMCRTA-detected nighttime thermal anomaly results in the continental United States. (a1)–(a4) Nighttime fire scene (15 × 15 km), Image central coordinates: 121.07 W, 46.761 N; Date: August 31, 2021. (b1)–(b4) Nighttime industrial heat source scene (15 × 15 km), Image central coordinates: 118.336 W, 33.835 N; Date: October 9, 2022.

Landsat 8 images (one nighttime fire and one nighttime industrial heat source, as shown in Fig. 16). Fig. 16(a1) shows a nighttime fire, with a large-area burning in a red ellipse and sporadic fire points in the green ellipse. The constructed reference temperatures (Fig. 16(a2) and (a3)] are similar to the observed temperatures [Fig. 16(a1)]. Both large-area and sporadic fires can be effectively identified by NMCRTA, as shown in Fig. 16(a4). For the industrial heat source in the red ellipse of Fig. 16(b1), NMCRTA can identify these nighttime thermal anomalies [Fig. 16(b4)] using two constructed reference temperatures [Fig. 16 (b2) and (b3)]. Therefore, the NMCRTA is effective in detecting nighttime fire and industrial heat sources, highlighting its good potential for all-day land surface anomaly detection.

D. Limitations of the NMCRTA Method

The NMCRTA method has four main limitations.

1) The application of NMCRTA is based on the basis of on-orbit rapid LST estimation. Currently, many on-orbit cloud detection and masking algorithms have been proposed. For example, Aybar et al. [51] proposed the Sentinel-2 network for on-orbit cloud detection and atmospheric correction based on lightweight evolutionary neural networks. However, on-orbit LST retrieval remains at the experimental stage, which poses challenges for accurately obtaining land surface emissivity and atmospheric information. Both cloud masking and LST estimation are expected to be achieved in the near future with the rapid development of on-orbit computing and storage capabilities [52].

2) In the NMCRTA, the fit reference LST is constructed using the previous year's LST data to characterize the normal thermal conditions of the land surface. For the coalfield fire pixels that persist for multiple years, the differences between fit reference LST and observed LST are small, which may be considered as normal pixels in the NMCRTA. It will lead to underidentification to some extent, but this situation is relatively rare due to the short duration of most thermal anomaly events.

3) The neighborhood median replacement algorithm is crucial for removing the influence of atmospheric conditions ($\Delta T_{\mathrm{atm}(x,y)}$). When more than half of the pixels in the $k \times k$ neighborhood are thermal anomalies (i.e., large-area thermal anomalies), the median value of temperature difference (Median($T_{\mathrm{sat}(x\pm k,y\pm k)} - T_{\mathrm{solarangle}(x\pm k,y\pm k)}$)) cannot represent the atmospheric conditions. Therefore, under these circumstances, the central pixel experiencing a thermal anomaly cannot be correctly detected using this median value. 4) Some land surface thermal anomaly events exhibit negative temperature variation instead of positive, such as avalanches. NMCRTA cannot detect this kind of thermal anomaly since its thresholds (T_α and T_β) are positive values. This study focuses only on positive thermal anomalies, since it is difficult to distinguish the negative temperature anomalies from ground shadows.

V. Conclusion

As a complement to the optical on-orbit detection method for land surface anomalies, a novel thermal infrared method (i.e., NMCRTA) was proposed in this study. NMCRTA could detect daytime and nighttime thermal anomalies, which enhances the applicability of on-orbit remote sensing detection techniques. It includes constructing reference temperatures (fit reference and real reference), calculating the temperature differences (including the step of removing atmospheric effects) between the observed LST and the two reference LSTs, and detecting thermal anomalies using corresponding thresholds. It performs well at the single-temporal regional scale, single-temporal scene scale, and continuous-time object scale. Two main conclusions are drawn as follows.

1) The validation results indicated that in the 100 × 100 km complex land-cover region, the NMCRTA can effectively detect deforestation, newly constructed buildings, and river drying with the $F1$-Score of 0.838. Furthermore, the NMCRTA exhibits excellent performance in detecting fires, deforestation, and landslides at the 15 × 15 km scene scale, with corresponding $F1$-Scores of 0.943, 0.857, and 0.791, respectively.

2) The NMCRTA performed well in object-scale temporal sensitivity analysis, which could continuously capture different thermal anomaly events associated with a newly constructed industrial heat source (e.g., thermal anomalies caused by deforestation for building a factory and operation as an industrial heat source).

The NMCRTA is promising for future real-time on-orbit remote sensing detection of daytime and nighttime land surface anomalies for two advantages.

1) The parameters (MAST, YAST, and dx) for the fit reference LST could be fit in advance using the previous year's LST dataset.

2) Once these parameters (three-band image) are stored in the satellite, rapid on-orbit thermal anomaly detection in single-temporal remote sensing images could be achieved. For that, the NMCRTA method will be further refined to address these three limitations: construction of a fit reference LST for multiple-year thermal anomalies, removal of the influence

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of atmospheric conditions for large-area thermal anomalies, and the detection of negative thermal anomalies. Currently, the proposed NMCRTA method cannot identify the specific types of thermal anomaly events. In future work, this limitation needs to be addressed by incorporating additional auxiliary information. For instance, the high-temperature fire-related anomalies may be identified by jointly analyzing the spectral characteristics of mid-infrared and thermal infrared bands, and the different types of low-temperature thermal anomalies may be distinguished through integrating optical remote sensing indices (such as NDVI, NDBI, or BSI). In this study, the visual interpretation of optical imagery for small-area anomaly events may be influenced by subjective judgment. The detection results could be further verified through on-site investigations in future studies.

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